Chapter 5.2
Mappings

Geometrical Representation:

To draw curve of complex variable (x, y) we take two axes i.e., one real axis and the other imaginary axis. A number of points (x, y) are plotted on z-plane, by taking different value of z (different value of x and y). The curve C is drawn by joining the plotted points. The diagram obtained is called **Argand diagram.**

Transformation:

For every point (x, y) in the z-plane, the relation $w = f(z)$ defines a corresponding point (u, v) in the w-plane. We call this "transformation or mapping of z-plane into w-plane". If a point z_0 maps into the point w_0, $w_0$is known as the image of z_0.

If the point $P(x, y)$ moves along a curve C in z-plane, the point $P'(u, v)$ will move along a corresponding curve C_1 in the w-plane. We, then, say that a curve C in the z-plane is mapped into the corresponding curve C_1 in the w- plane by the relation $w = f(z)$.

Translation, Rotation and reflection are the standard transformations. Terms such as **translation, rotation** and **reflection** are used to convey dominant geometric characteristics of certain mappings.

Translation

$w = z + C$,

where,

$$C = a + ib$$
$$z = x + iy$$
$$w = u + iv$$

Hence, $u + iv = x + iy + a + ib$

So, $u = x + a$ and $v = y + b$

$x = u - a$ and $y = v - b$

On substituting the values of x and y in the equation of the curve to be transformed, we get the equation of the image in the w-pane.

As an example, the mapping $w = z + 1$ where $z = x + iy$, can be thought of as a translation of each point of z one unit to the right.

Example: 5.2.1

Let the rectangular region R in z-plane which is bounded by the lines

$$x = 0, y = 0, x = 2, y = 1.$$

Determine the region R' of the w-plane into which R is mapped under the transformation $w = z + 1$.

Solution:

Given $w = z + 1$

or, $u + iv = (x + 1) + iy$.

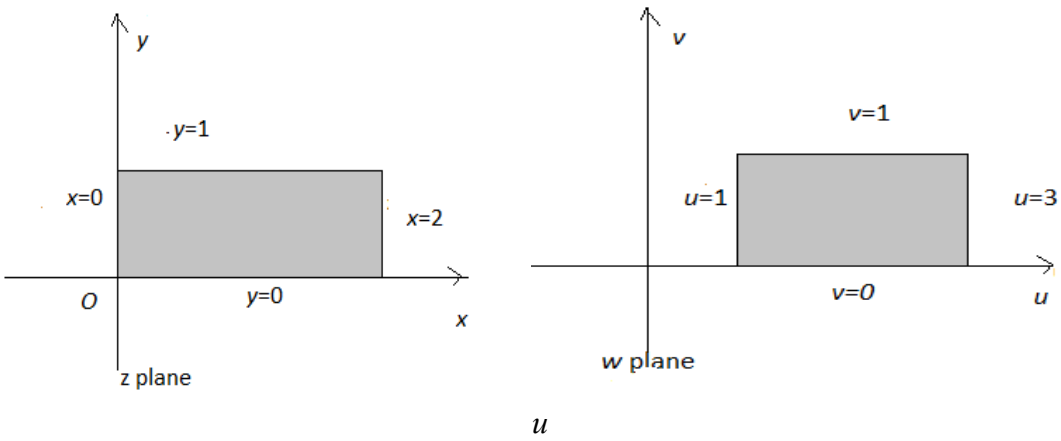
Hence $u = x + 1$ and $v = y$.

when $x = 0,\ u = 1,$

$y = 0,\ v = 0$

$x = 2,\ u = 3,$

$y = 1,\ v = 1$



u

Rotation:

The mapping $w = iz$ where $z = re^{i\theta}$ and $i = e^{i\frac{\pi}{2}}$, can be thought of as a rotation of the radius vector for each non-zero point z through a right angle about the origin in the counterclockwise direction.

Example: 5.2.2

Let the rectangular region R in z-plane which is bounded by the lines

$$x = 0, y = 0, x = 2, y = 1.$$

Determine the region R' of the w-plane into which R is mapped under the transformation

$$w = iz.$$

Solution:

Given $w = iz$

or, $u + iv = -y + ix.$

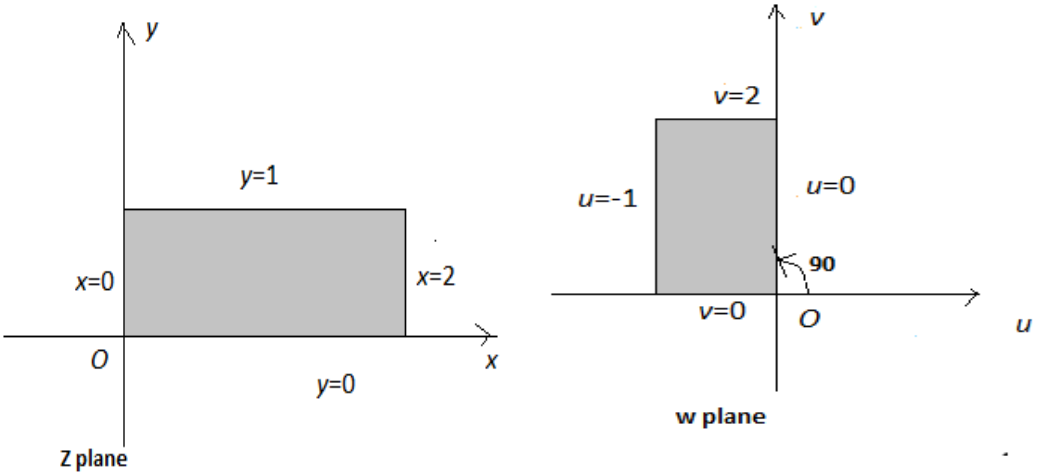
Hence $u = -y$ and $v = x.$

when $x = 0,\ v = 0,$

$y = 0,\ u = 0$

$x = 2,\ v = 2,$

$y = 1,\ u = -1$



Reflection:

The mapping $w = \bar{z}$ transforms each point of $z = x + iy$ into its reflection in the real axis.

Example: 5.2.3

Let the rectangular region R in z-plane which is bounded by the lines

$$x = 0, y = 0, x = 2, y = 1.$$

Determine the region R' of the *w*-plane into which R is mapped under the transformation

$$w = \bar{z}\ .$$

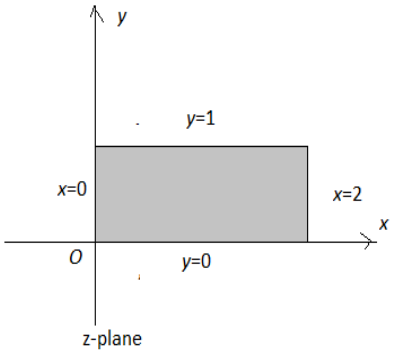
Solution:

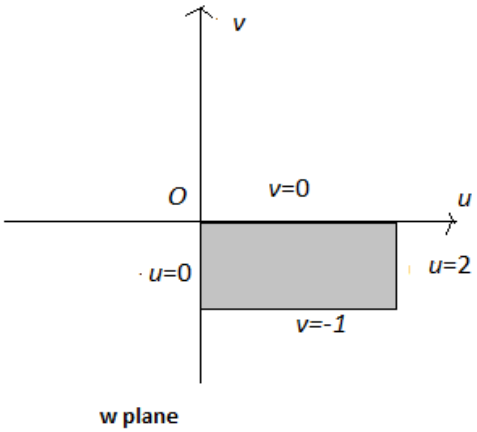
Given $w = \bar{z}$

or, $u + iv = x - iy$.

Hence $u = x$ and $v = -y$.

when $x = 0,\ u = 0,$
$y = 0,\ v = 0$
$x = 2,\ u = 2,$
$y = 1,\ v = -1$





Example: 5.2.4

Given triangle T in the z-plane with vertices at $-1 + 2i, 1 - 2i$ and $1 + 2i$. Determine the triangle T' of the *w*-plane into which T is mapped under the transformation $w = \sqrt{2}e^{\frac{\pi i}{4}}z$.

Solution:

Given $w = \sqrt{2}e^{\frac{\pi i}{4}}z = (1 + i)(x + iy)$

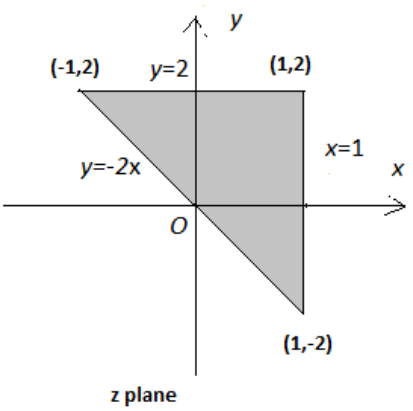
or, $u + iv = (x - y) + i(x + y)$.

Hence $u = x - y$ and $v = x + y$.

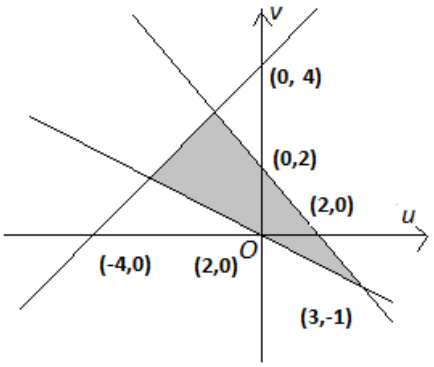
When
$x = 1,$
$y = 2$
$y = -2x$

The vertices of the triangle are $-1 + 2i, 1 - 2i, 1 + 2i$. Hence the sides are $x = 1,\ y = 2$ and $y = -2x$.

$u = 1 - y, v = 1 + y \quad => u + v = 2$
$u = x - 2, v = x + 2 \quad => u - v = -4$
$u = 3x, v = -x \quad => u + 3v = 0$



z plane



w plane

Exercise Set 5.2.1

1. Answer the following questions based on the topics studied above:

a. Let the rectangular region R in z-plane which is bounded by the lines $x = 2, y = 0, x = 5$ and $y = 4$. Determine the region R' of the w-plane into which R is mapped under the following transformations:

i. $w = 2z - (2 + 3i)$

ii. $w = \frac{1}{2}e^{\frac{\pi i}{2}}z + 2i$

iii. $w = \sqrt{2}e^{\frac{\pi i}{4}}z - (1 - i)$

iv. $w = e^{i\pi}z + 3 + i$

v. $w = \frac{1}{\sqrt{2}}e^{\frac{\pi i}{4}}z + 1 - 3i$

b. Given triangle T in the z-plane with vertices at $1, 1 - 3i$ and $3 - i$. Determine the triangle T' of the w-plane into which T is mapped under the following transformations:

(i) $w = 3z + 1 - 3i$, (ii) $w = iz + 3 + 2i$,

(iii) $w = (1 + 2i)z - i$, (iv) $w = \frac{1}{2}e^{\frac{\pi i}{2}}z - 4$.

c. A triangle in the z-plane is defined by the straight lines

$$y = 3x,\ y = -x \text{ and } x = -1.$$

Mapping of the triangle from z- plane to w-plane is defined by, $w = \frac{1}{\sqrt{2}}e^{-\frac{i\pi}{4}}z - i$.

Find the triangle in w-plane by answering the following question:

i. Find $u(x, y)$ and $v(x, y)$ so that, $w = u(x, y) + i\, v(x, y)$.

ii. Use the relations between x and y in the equation of the sides of the triangle in the z-plane to find the relationship between u and v in the w-plane.

iii. Draw the new triangle in w-plane and identify the vertices.

2. Problems from book (A first course in complex analysis with applications, Dennis G. Zill)

a. Find the image S' of the square S with vertices at $1 + i, 2 + i, 2 + 2i$, and $1 + 2i$ under the linear mapping $T(z) = z + 2 - i$

b. Find the image of the rectangle with vertices $-1+i, 1+i, 1+2i$, and $-1+2i$ under the linear mapping $f(z) = 4i \cdot z + 2 + 3i$
c. Find the image of the triangle with vertices 0, 1, and i under the given linear mapping $w = f(z)$:

i. $f(z) = z + 3i$
ii. $f(z) = z + 2 - i$
iii. $f(z) = 3iz$
iv. $f(z) = 1 + iz$
v. $f(z) = 2z - i$
vi. $f(z) = 4 - 3iz + 1 - 3i$
vii. $f(z) = \frac{1}{\sqrt{2}} e^{\frac{i\pi}{4}} z + \frac{1}{2} - \frac{i}{2}$
viii. $f(z) = \sqrt{2} e^{i\frac{\pi}{4}} z + 2i$
ix. $f(z) = -2z + 1 - \sqrt{3}i$
x. $f(z) = (3 - 2i)z + 1$